

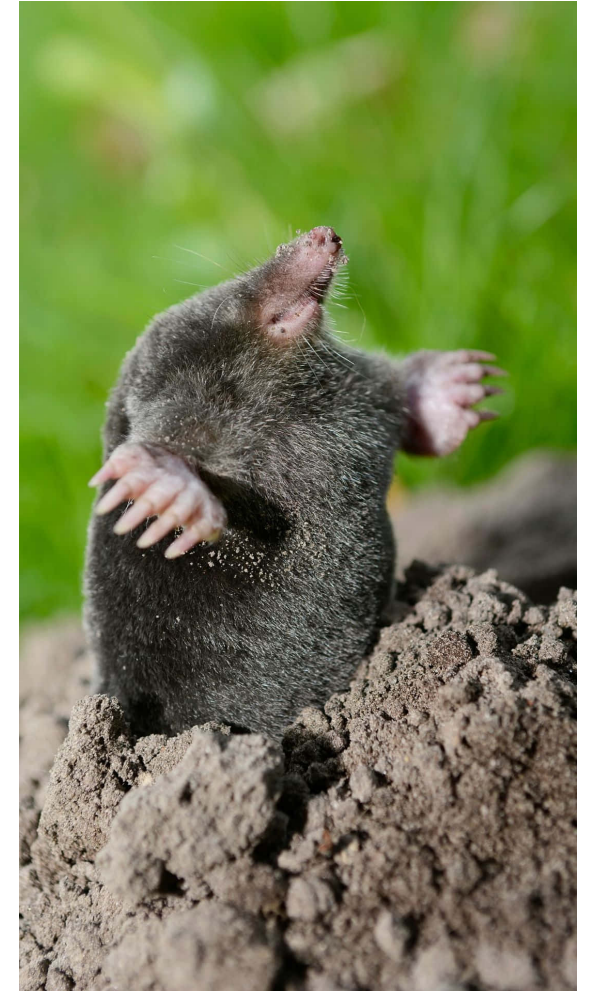
Welcome to SQLQ, Lesson 1!

Central SQ Slide Deck

Last Updated September 4, 2026 by Percy Smith

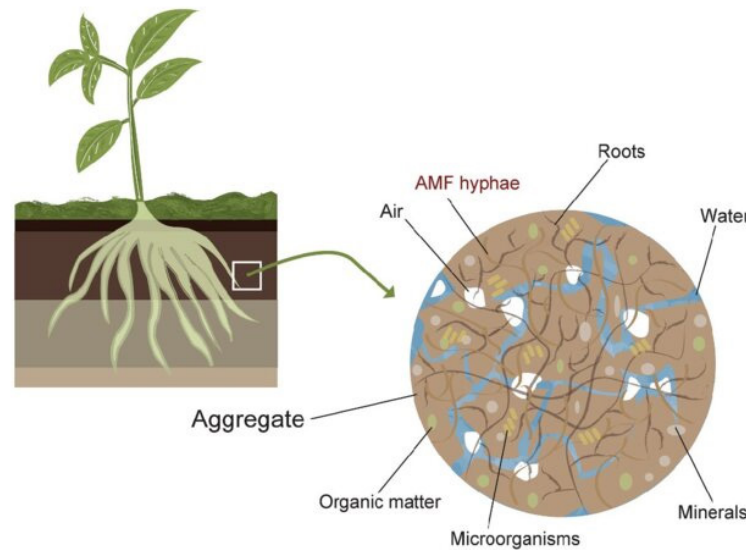


What Actually is Soil?



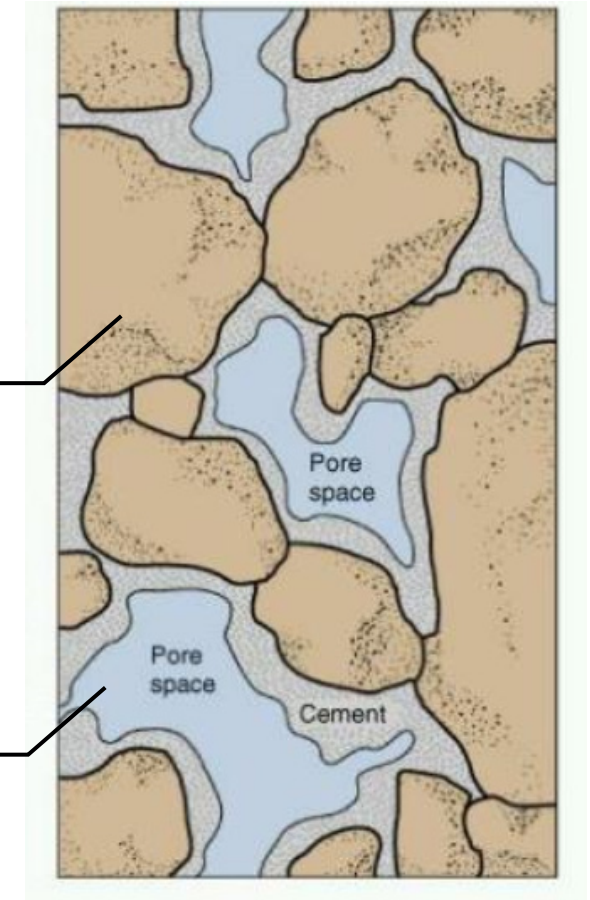
Soil is Made Of...

- Solids
 - Organic Matter (Living & Dead) (0-5%)
 - Minerals (45-50%)
- Pore Space
 - Liquid
 - Gas/Air



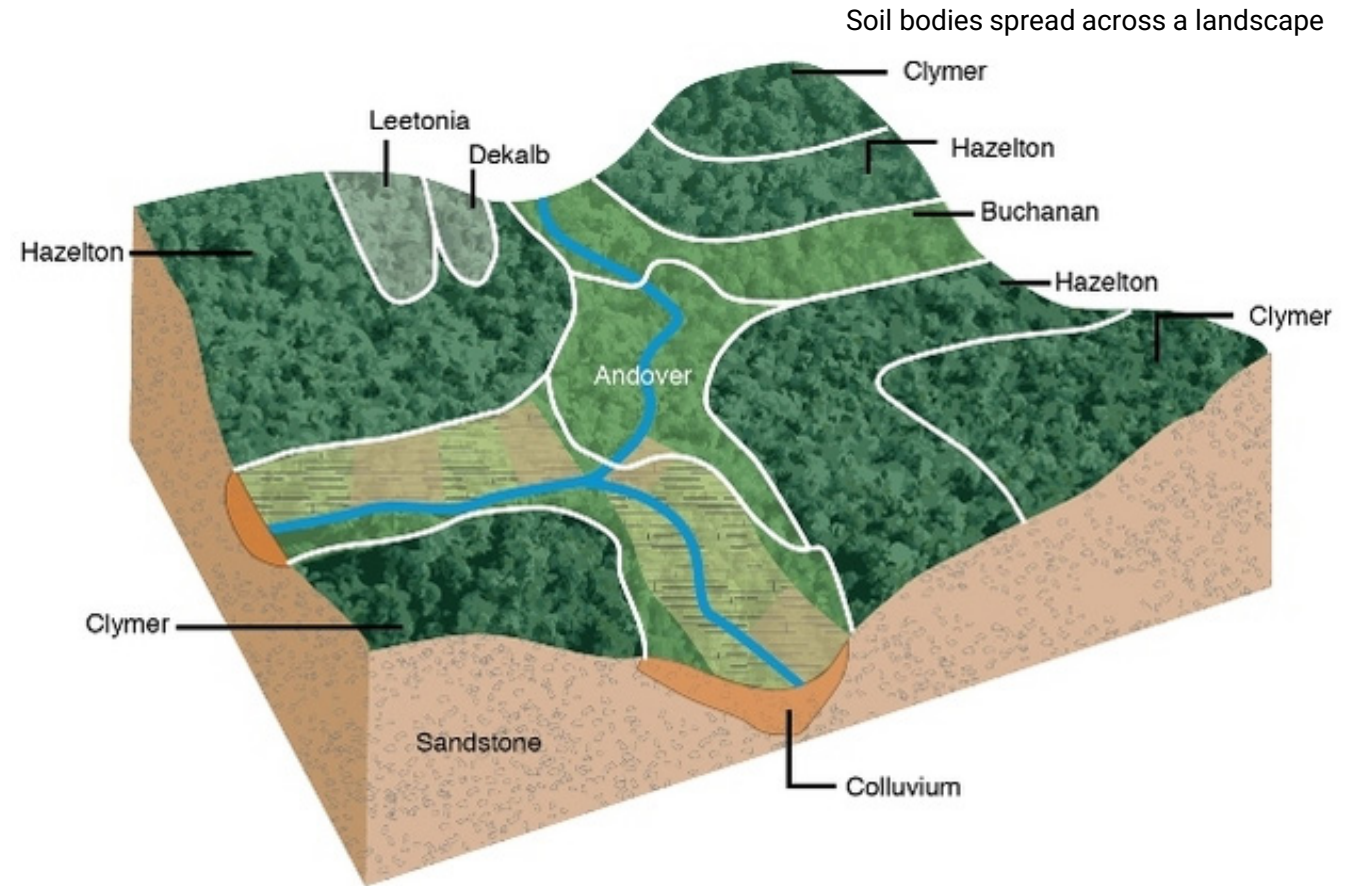
Solids (50%)

Pore Space (50%)



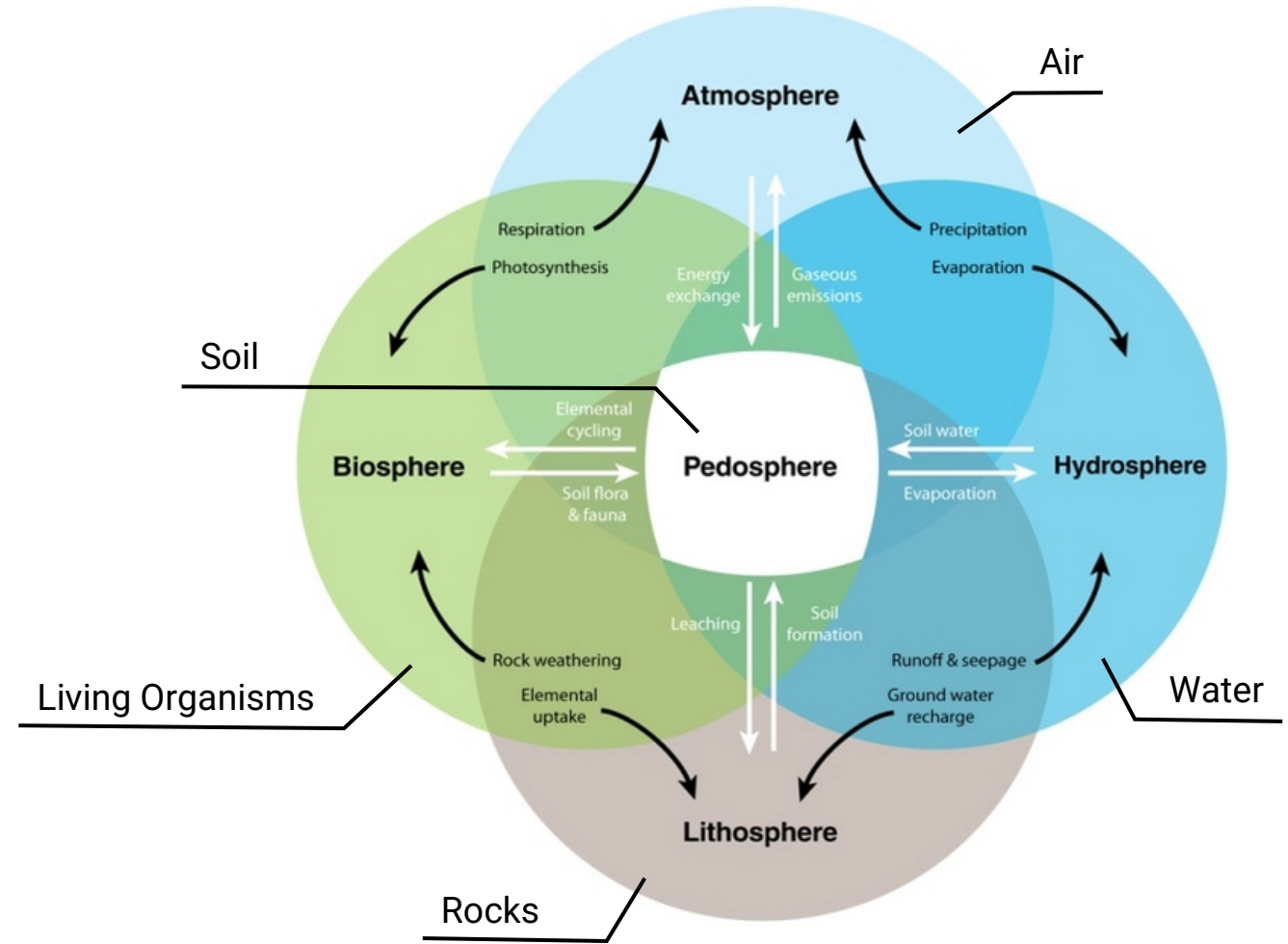
Soil is a Nexus

- Soils are huge!
 - Soil bodies are natural bodies with size, form and history

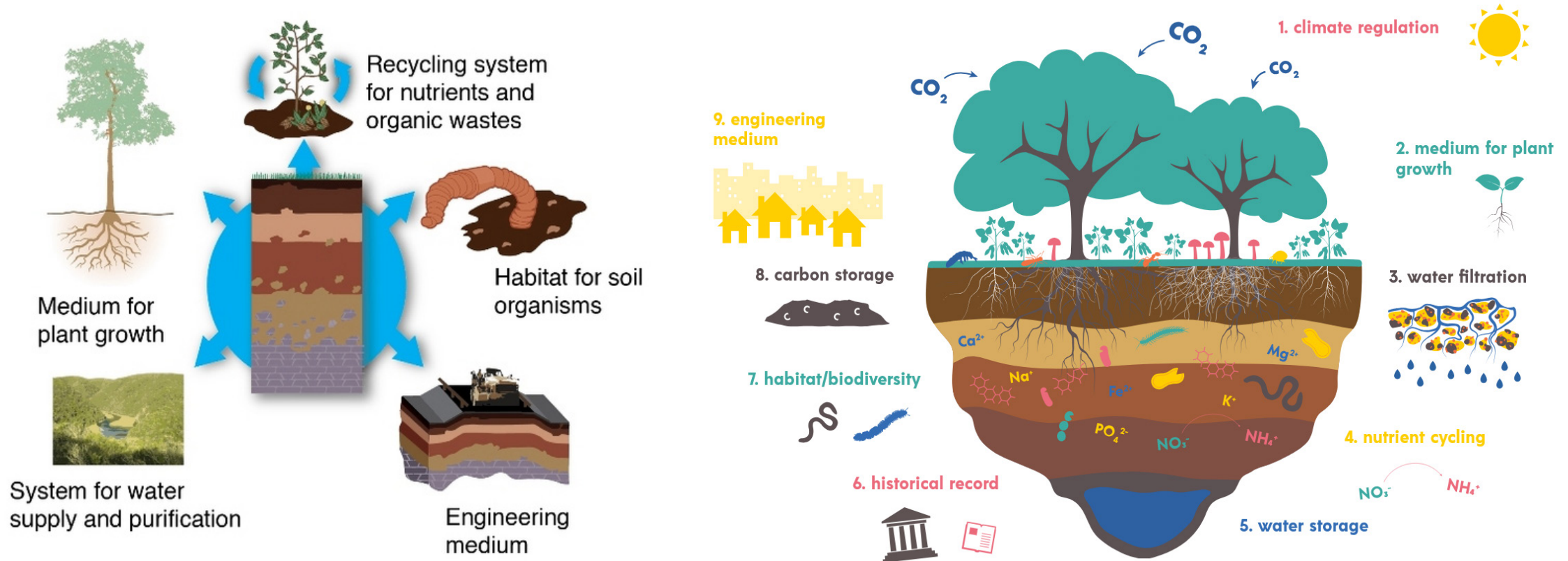


Soil is a Nexus

- Soils are huge!
 - Soil bodies are natural bodies with size, form and history
- Soil bodies are a nexus point for rock, water, air, and living organisms
 - Soil systems provide functions greater than the sum of their parts



Soils Provide Critical Services



How Can We Describe & Measure Soil?



How do we group, classify, and taxonomize soil?

What are characteristics of this soil?

What capacity does this soil have for this ecosystem?

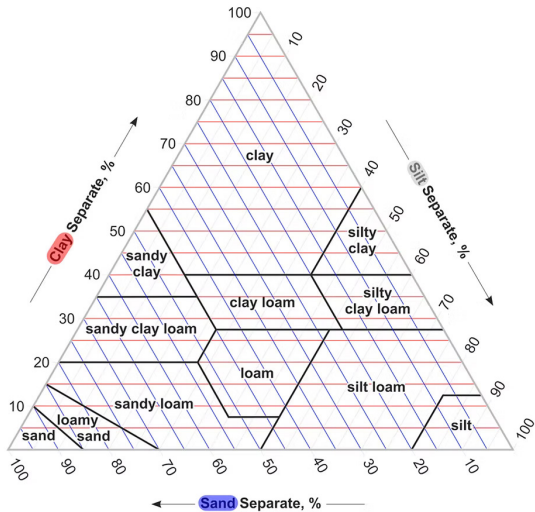
Soil Capability

Properties that relate to the soil's composition; these often go unchanged

Inherent Soil Qualities



Formation Factors

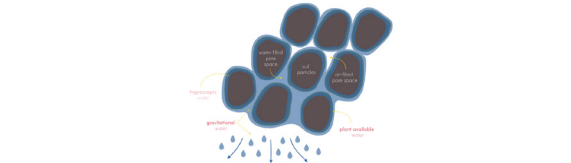


Soil Texture

Soil Health

Properties that change with use, management, and the environment

Dynamic Soil Qualities

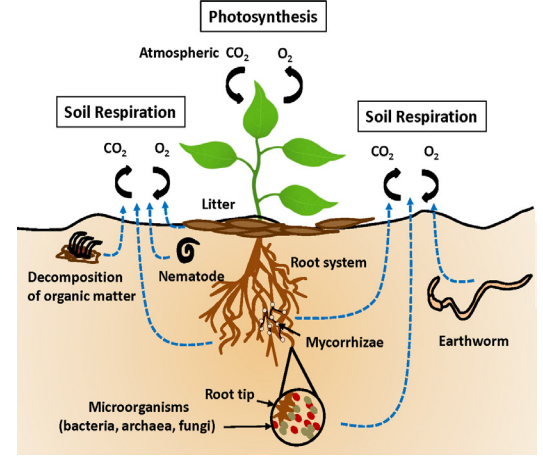


Soil Moisture

Is the soil able to function at its "full capacity"?

Can the soil function as a living ecosystem?

Can the soil sustain animals, plants, and humans?



Soil Respiration



How is Soil Formed?



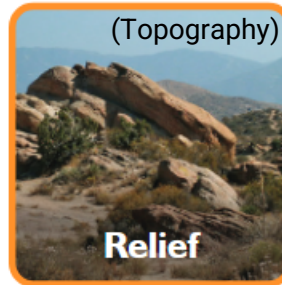
CLimate

CL—Temperature and moisture influence the speed of chemical reactions, which in turn help control how fast rocks weather and dead organisms decompose. Soils develop fastest in warm, moist climates and slowest in cold or arid (dry) ones.



Organisms

O—Plant roots spread, animals burrow, and bacteria eat. These and other soil organisms speed up the break down of large soil particles into smaller ones. Roots are a powerful soil-forming force, cracking rocks as they grow. And roots produce carbon dioxide that mixes with water, forming an acid that wears away rock.



(Topography)

Relief

R—The shape of the land and the direction it faces make a difference in how much sunlight a soil gets and how much water it holds. Deeper soils form at the bottom of a hill than at the top because gravity and water move soil particles down the slope.



Parent material

P—Just like you inherited characteristics from your parents, every soil inherits traits from the materials in which it forms. Soils that formed from limestone bedrock, for example, are rich in calcium. Soils that formed from materials at the bottom of lakes are high in clay.



Time

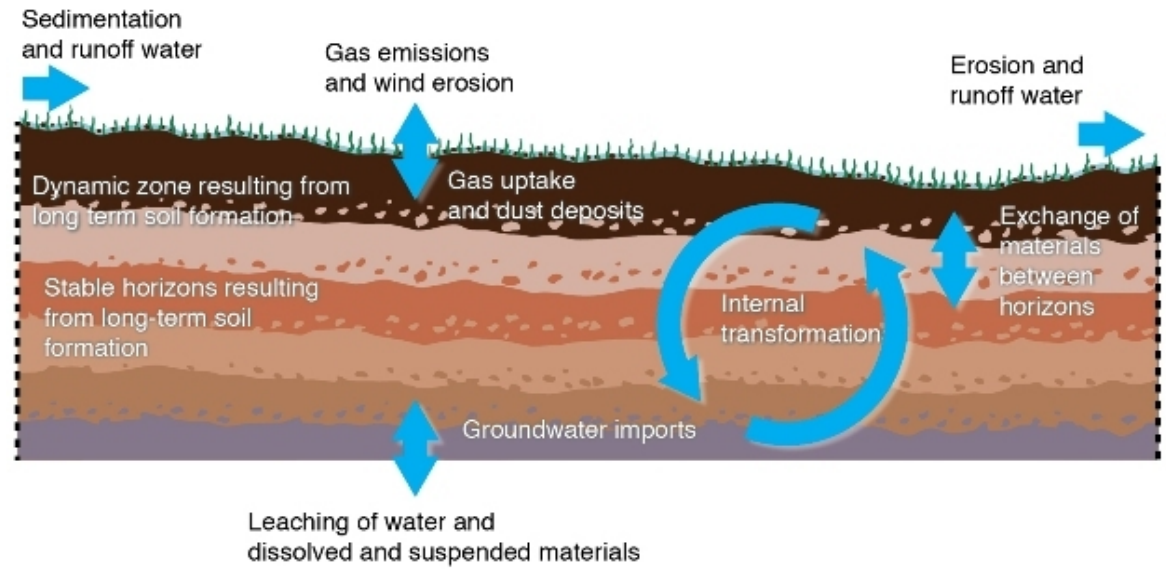
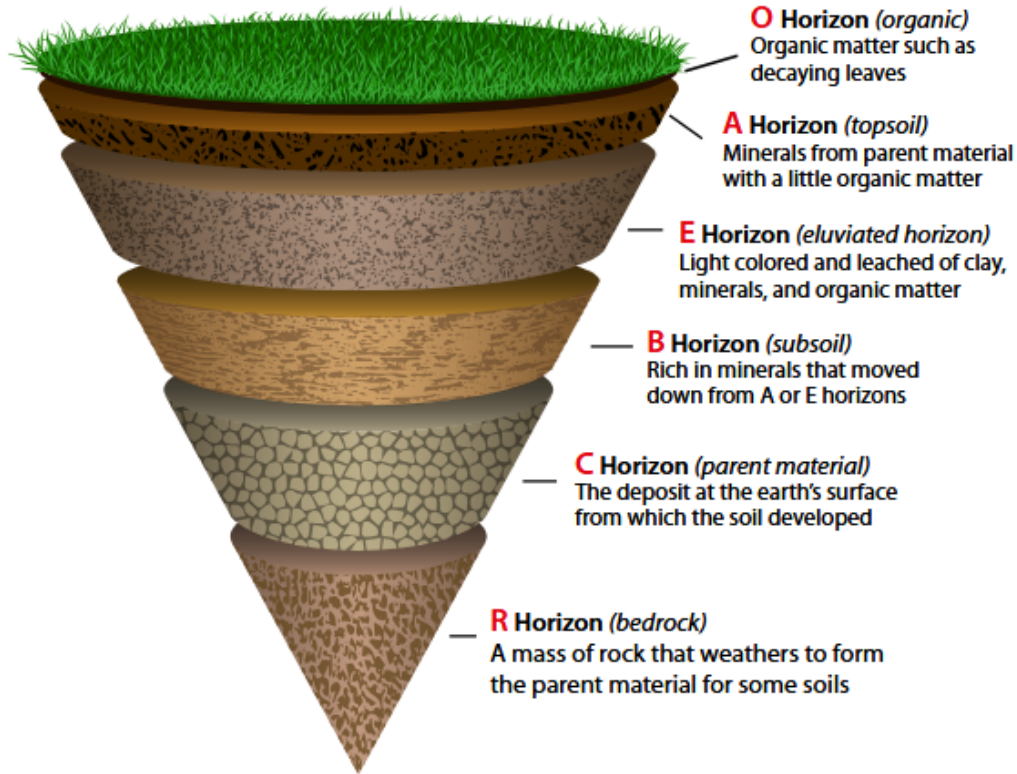
T—Older soils differ from younger soils because they have had longer to develop. In the northern US, soils tend to be young because glaciers covered the surface during the last Ice Age. In the southern US, there were no glaciers, so the soils have been exposed for a longer time, making them more weathered.

Soils are different because of their formation!

There are 5 major factors influencing soil formation, or **CLORPT** for short!



Aging Soil



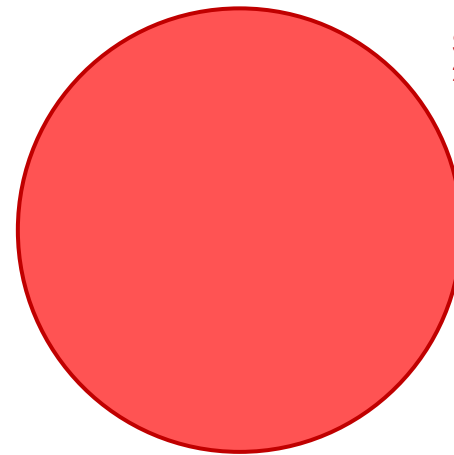
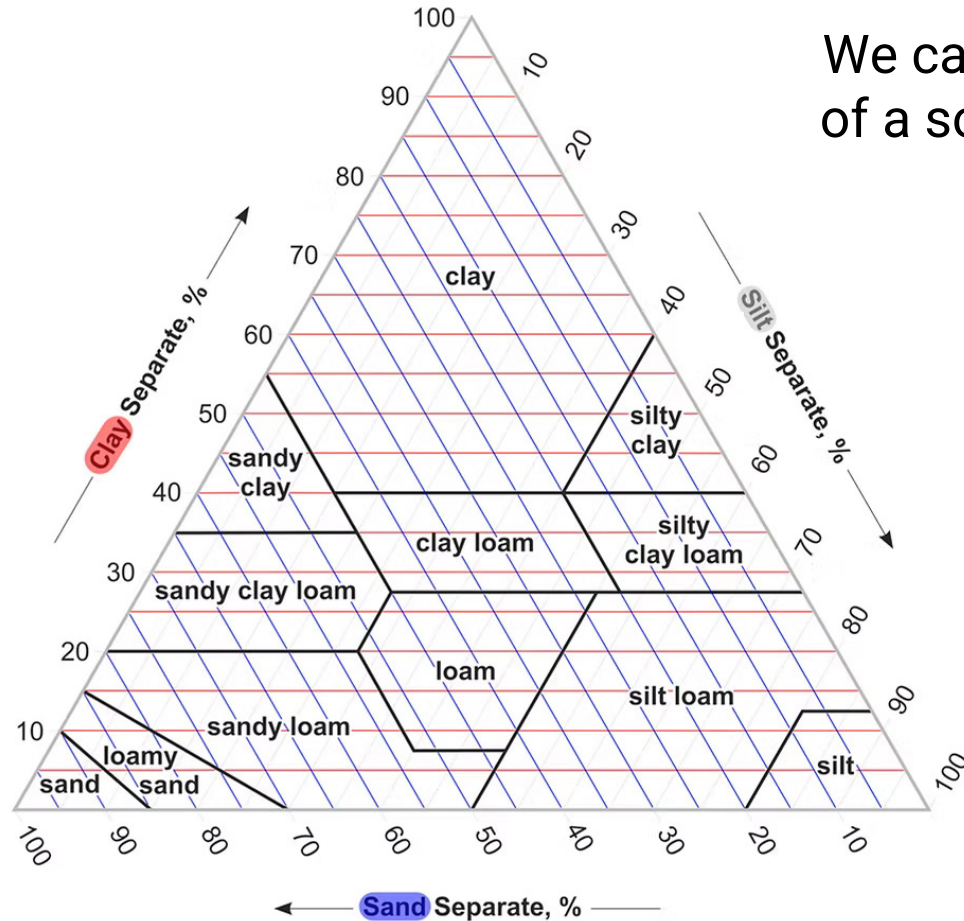
As soils age, they form **Horizons**, or layers in the soil

These Horizons come together to form a **Soil Profile**



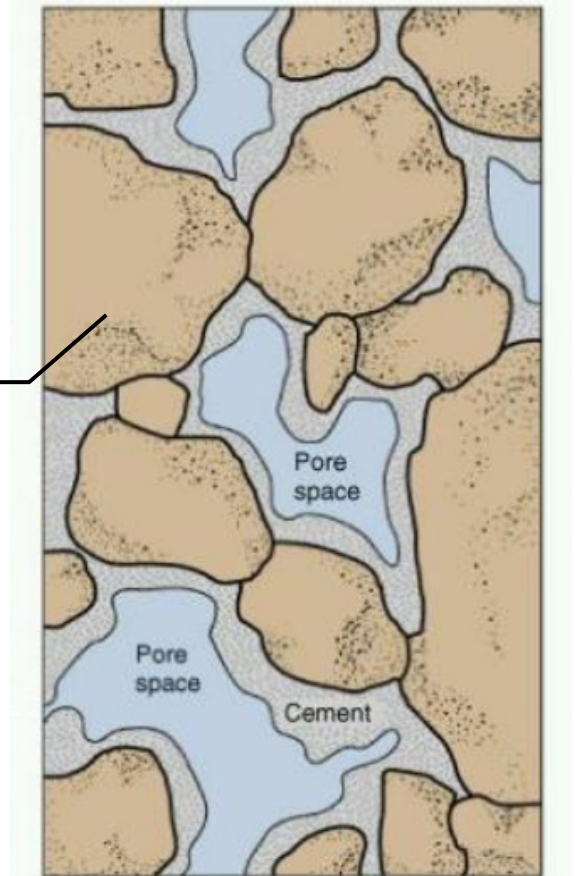
Soil Texture

We can determine the texture of a soil by looking at the size of mineral grains!

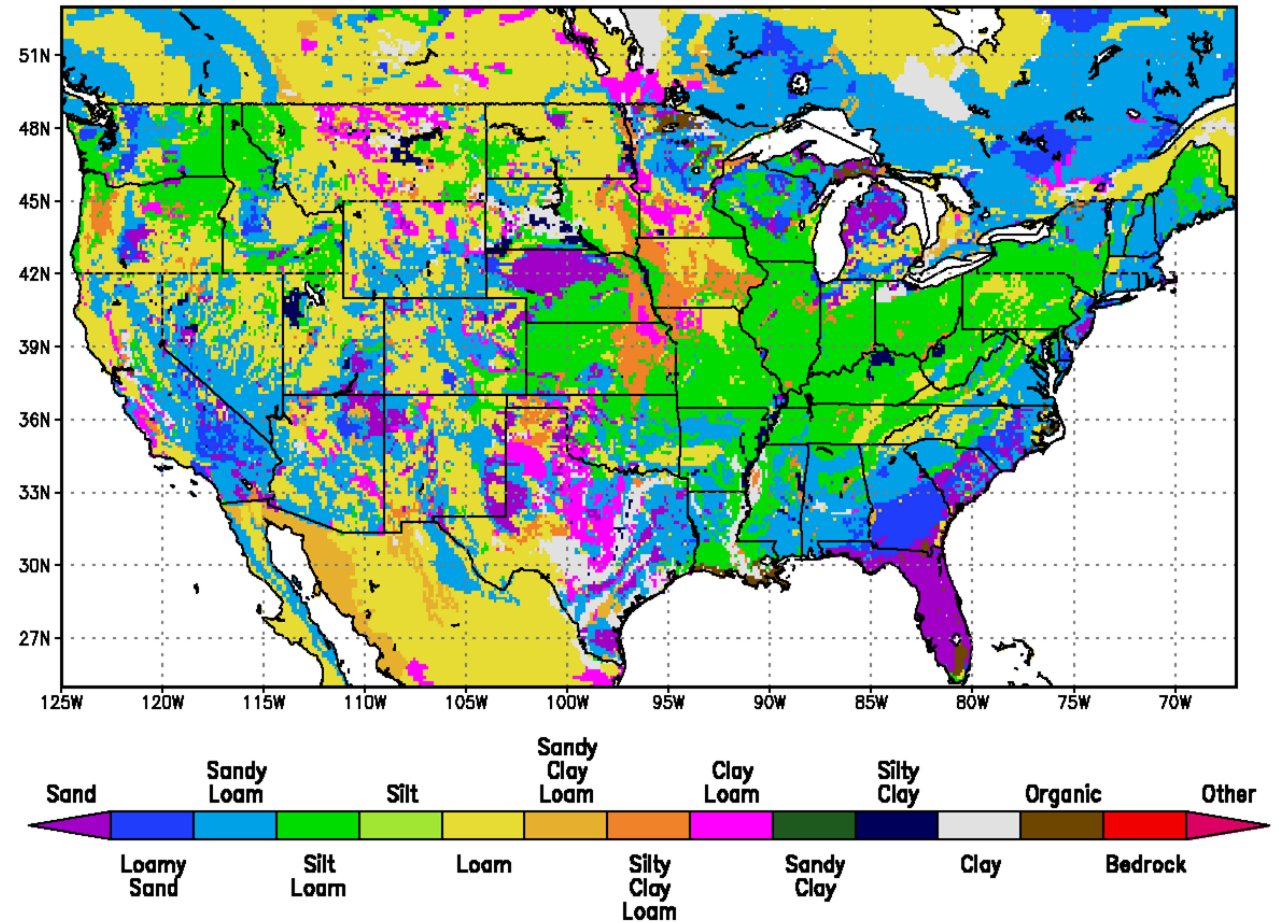
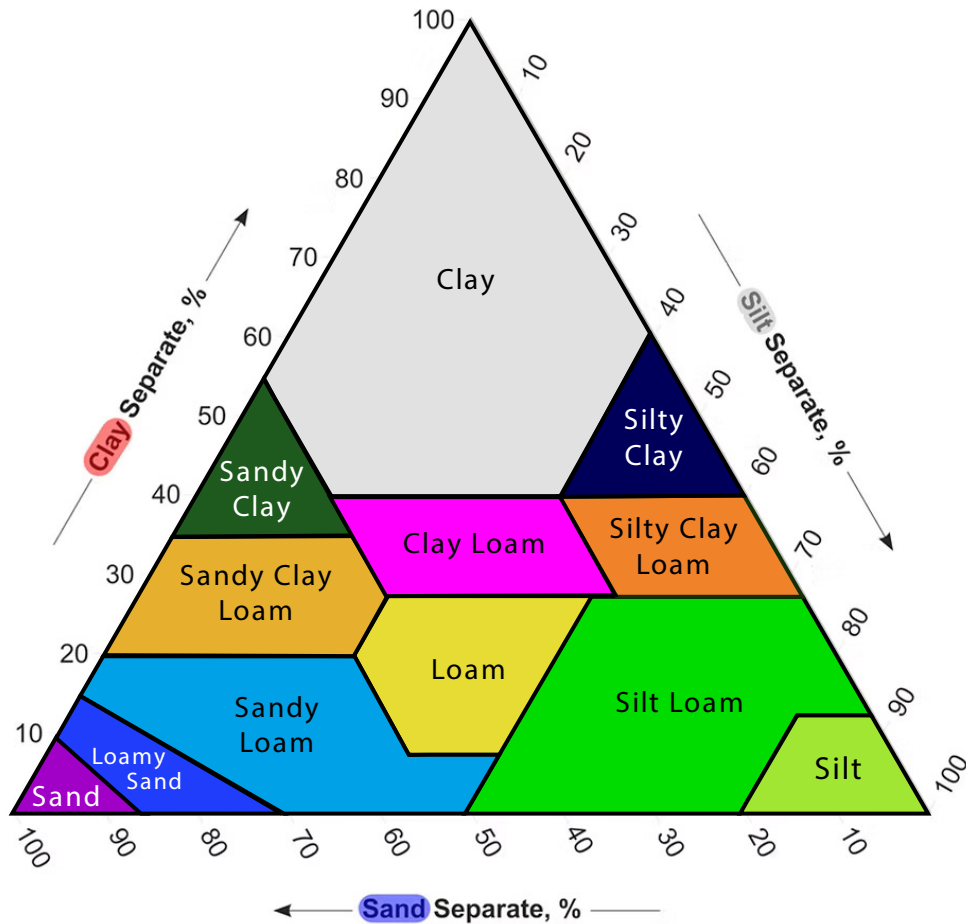


- Sand
2.00-0.05 mm
- Silt
0.05-0.002 mm
- Clay
<0.002 mm

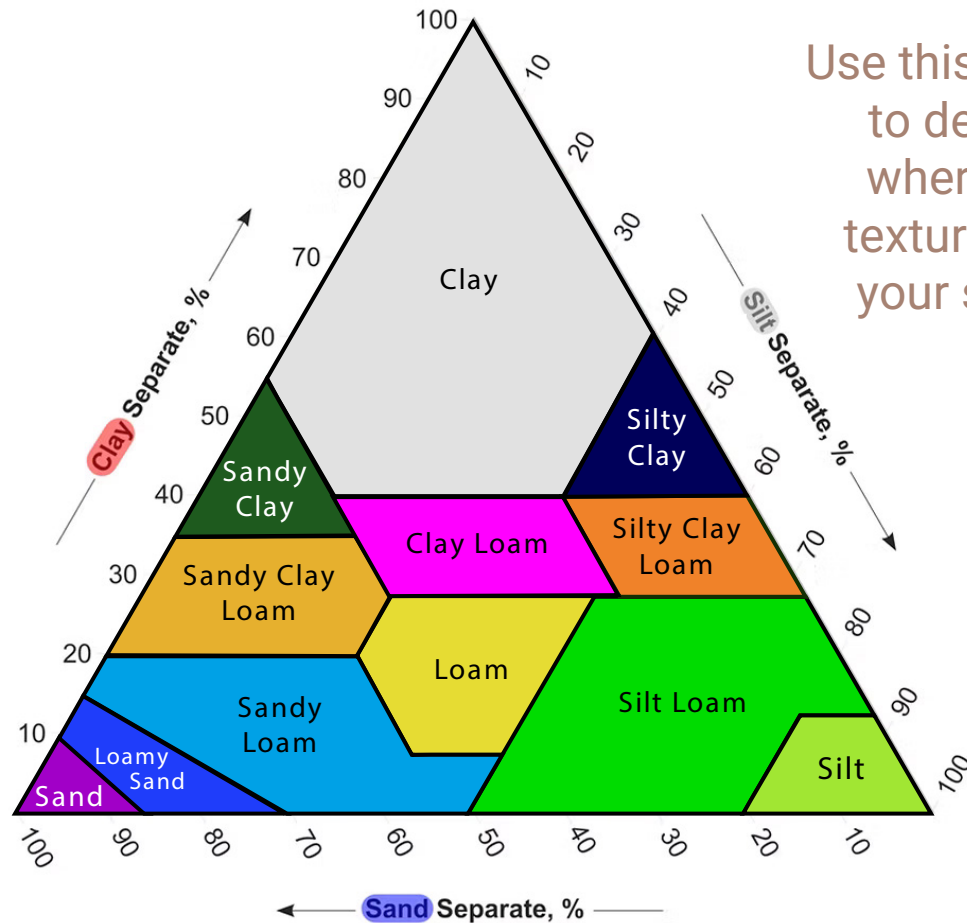
Solids (50%)



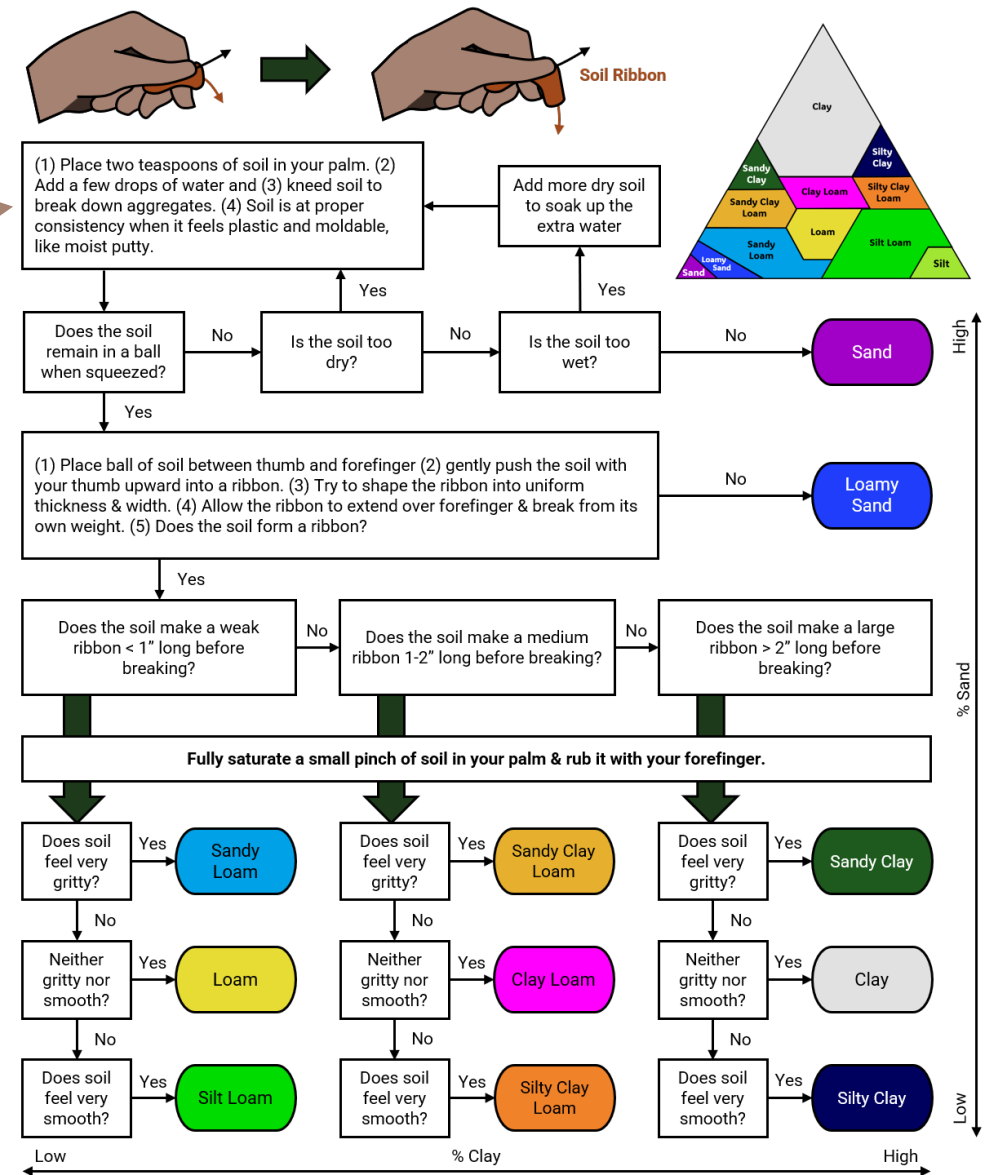
Soil Texture in the US!



Measuring Soil Texture!



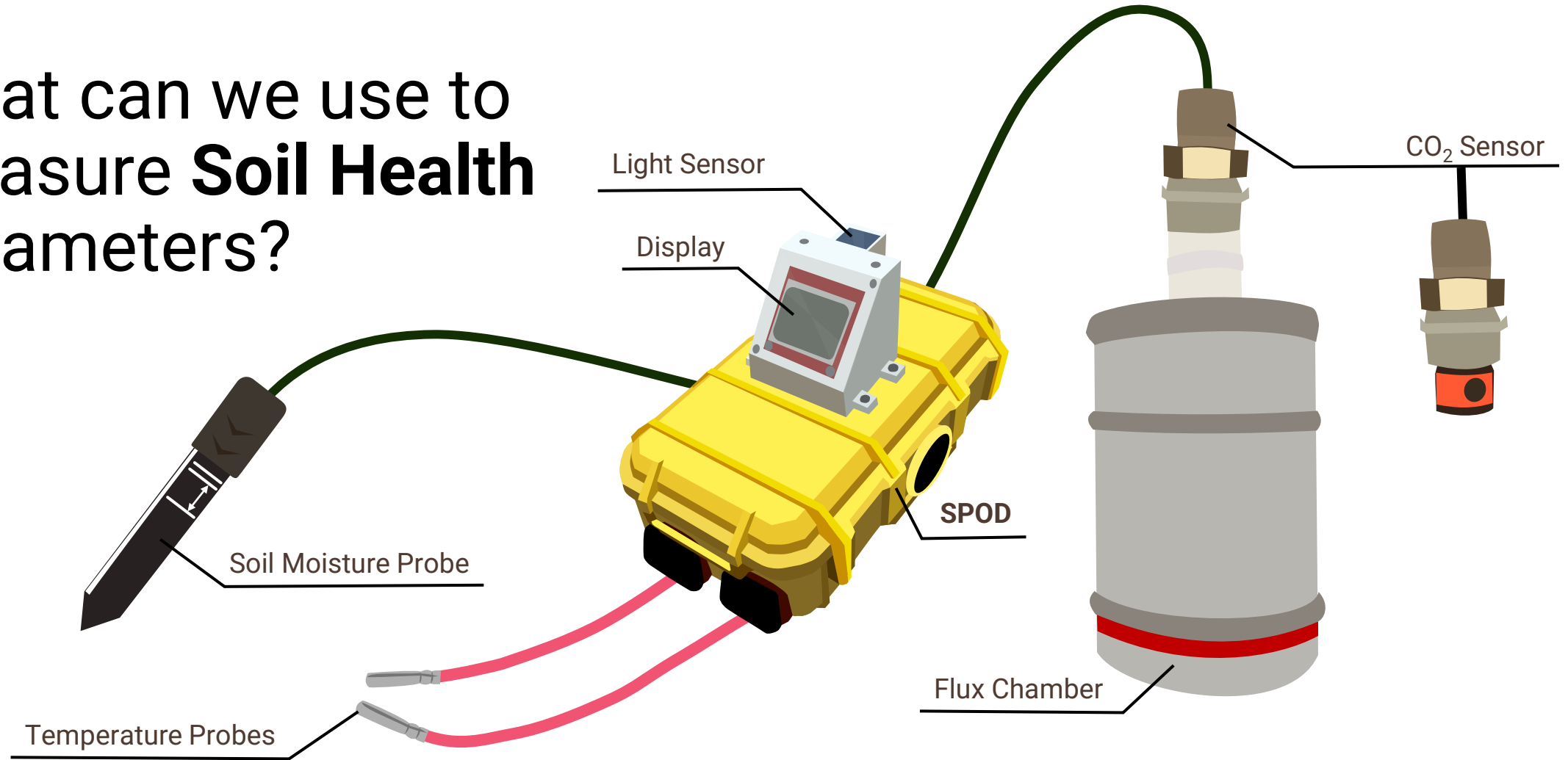
Use this flow chart to determine where on the texture triangle your soil falls!



Based on Key Resources from:
Agriculture NSW Water Unit. (2015). Determining soil texture using the ribboning technique (Primefact 1363). NSW Department of Primary Industries.
Soil Texture By Feel Flow Chart. (2013). Oregon State University.
Last Updated: 08/31/2026 by Percy Smith

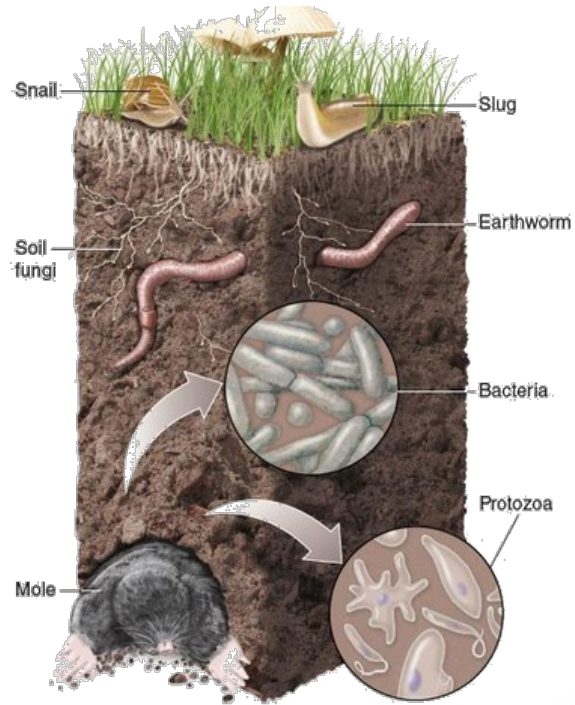


What can we use to measure **Soil Health** parameters?

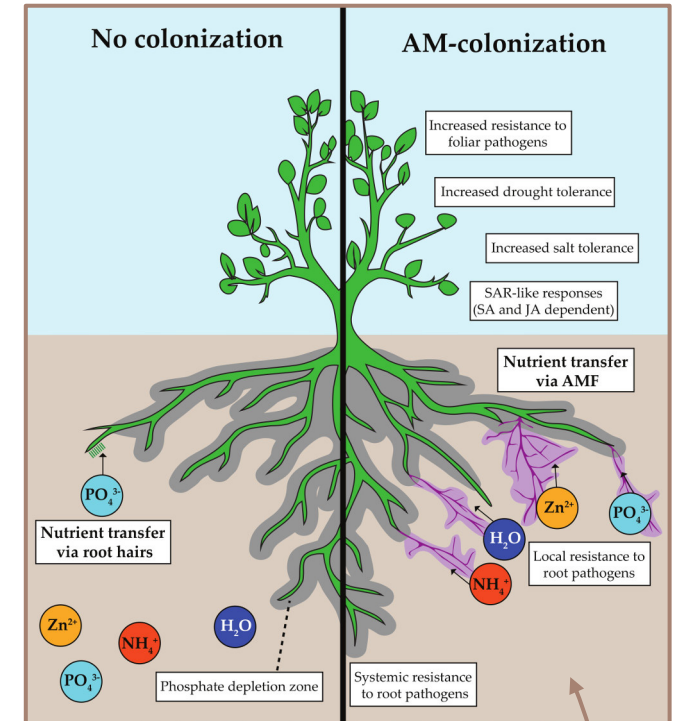
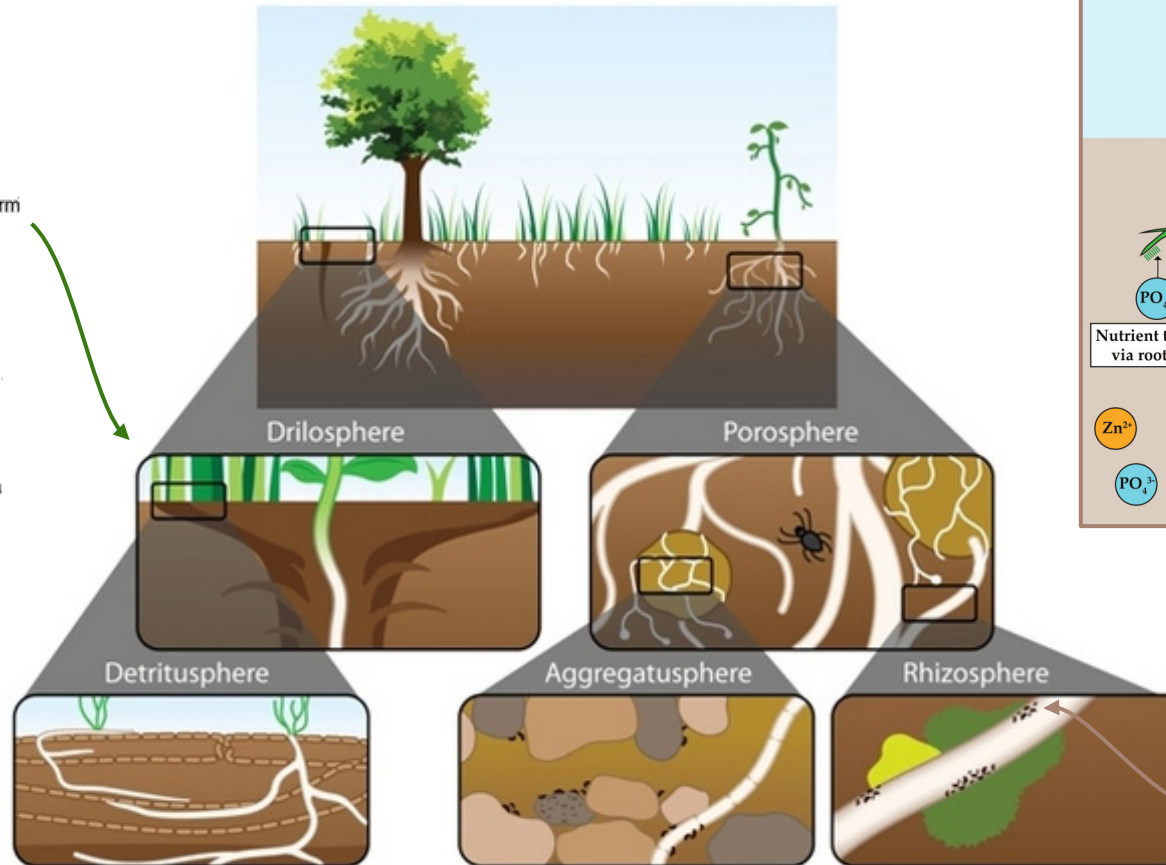


Soil is Full of Life!

AMF lives in the **Rhizosphere**, where it forms **symbiotic relationships** with terrestrial plants to help them succeed in their environment!



Earthworms are “regarded as ‘soil ecological engineers’” because their actions create soil & build the **Drilosphere**!



Engineering & Applied Science

UNIVERSITY OF COLORADO BOULDER

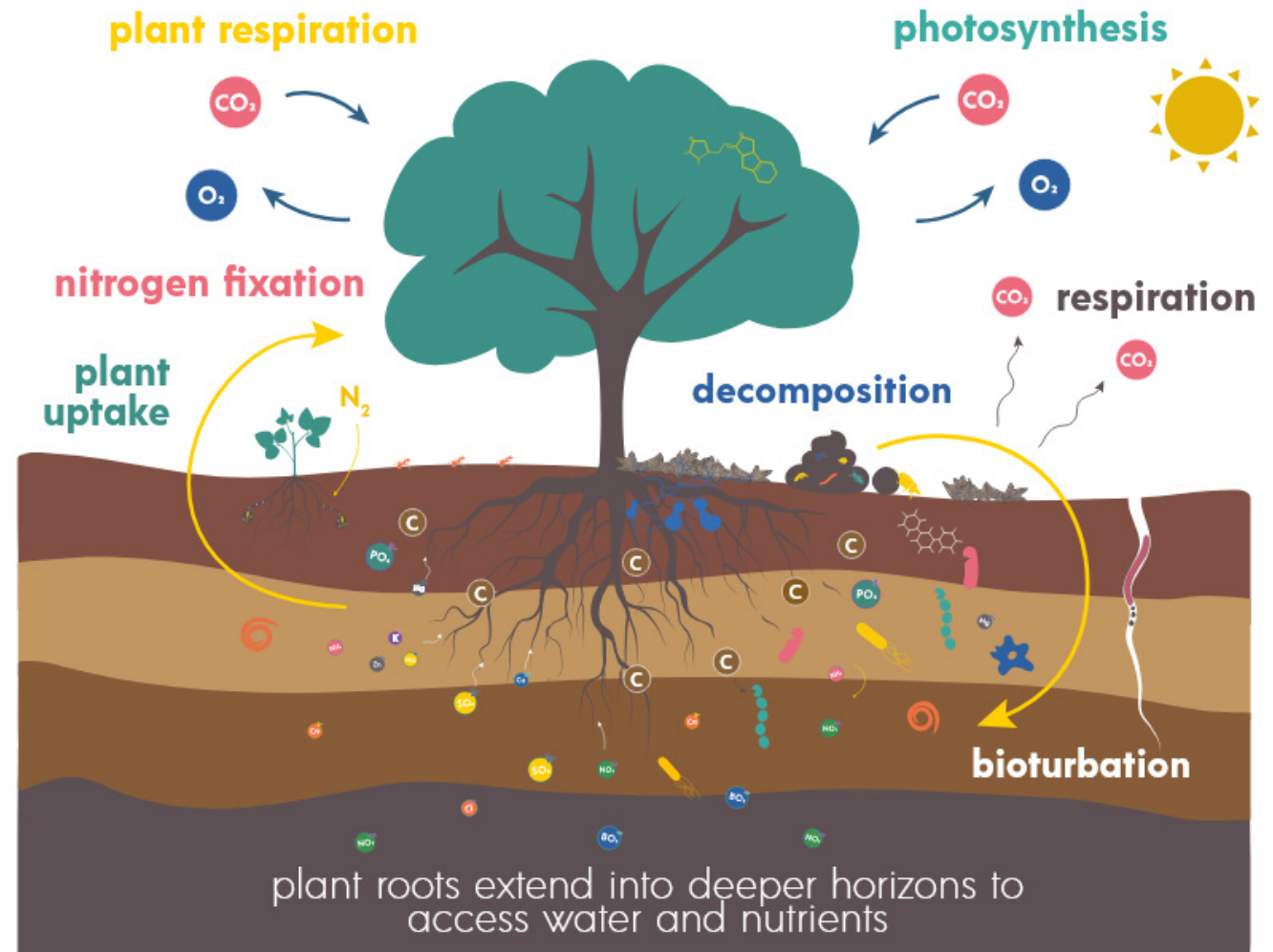
Huang, C., Wang, L., Wu, W.-M., Capowicz, Y., Qiao, Y., & Hou, D. (2025). When plastisphere and drilosphere meet: Earthworms facilitate microbiome and nutrient turnover to accelerate biodegradation of agricultural plastic films. *Environment International*, 196, 109309. <https://doi.org/10.1016/j.envint.2025.109309>
 Jacott, C. N., Murray, J. D., & Ridout, C. J. (2017). Trade-Offs in Arbuscular Mycorrhizal Symbiosis. *Agronomy*, 7(4), 75. <https://doi.org/10.3390/agronomy7040075>
 Friedland, A. J., & Relyea, R. (2015). *Environmental Science for AP* (Second Edition). W.H. Freeman and Company.
 Fortuna, A.-M. (2012). The Soil Biota. *Nature Education Knowledge*, 3(10), 1.



Soil Respiration

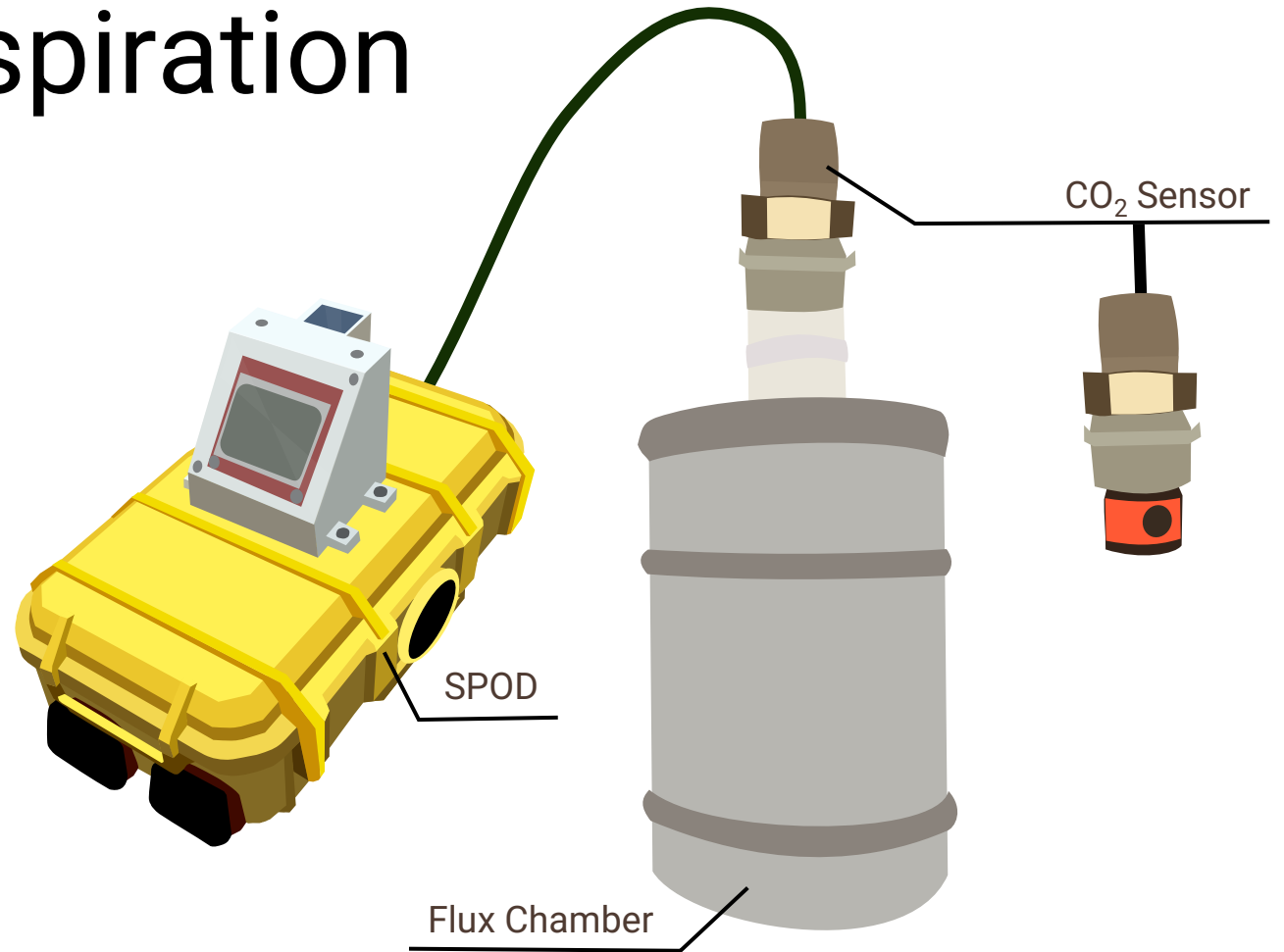
Because soil is so full of life, soil breathes!

Soil respiration is a measure of how much carbon dioxide is released from soil



Measuring Soil Respiration

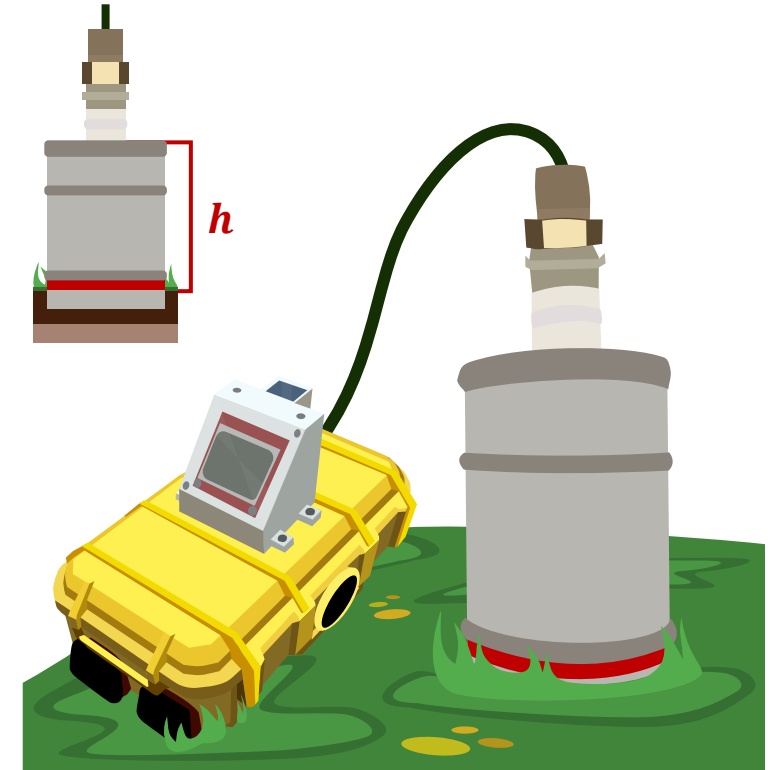
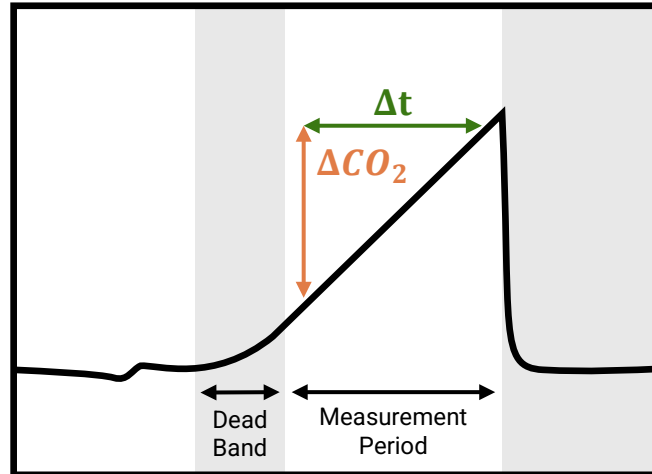
- We have two very different soil samples
 - One has a lot more opportunity to host life than the other!
- What do we expect to see when we measure soil respiration?



Calculating CO₂ Flux

Variable	Value	Units
P_{atmos}	76,500	Pa
R	8.314	$\frac{m^3 \cdot Pa}{K \cdot mol}$
M_{CO_2}	44	$\frac{g}{mol} CO_2$
h	15.75	cm
T_{avg}		K
ΔCO_2		ppm
Δt		s
Φ_{CO_2}		$\frac{kg CO_2}{m^2 \cdot s}$

$$\Phi_{CO_2} = \frac{\Delta CO_2}{\Delta t} M h \frac{P_{atmos}}{R T_{avg}}$$

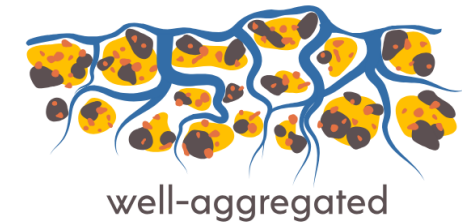
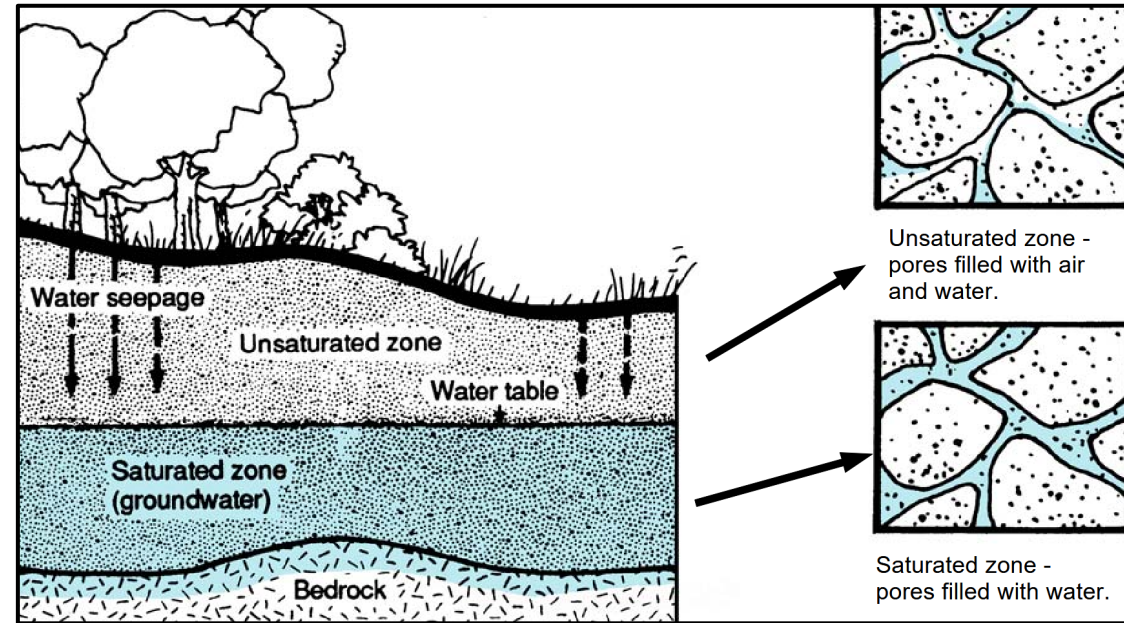


Soil Stores Water!

Soil can store water in its **pore space & organic matter**

This means that soil is critical for:

1. **Groundwater Storage**
2. **Flood Protection**

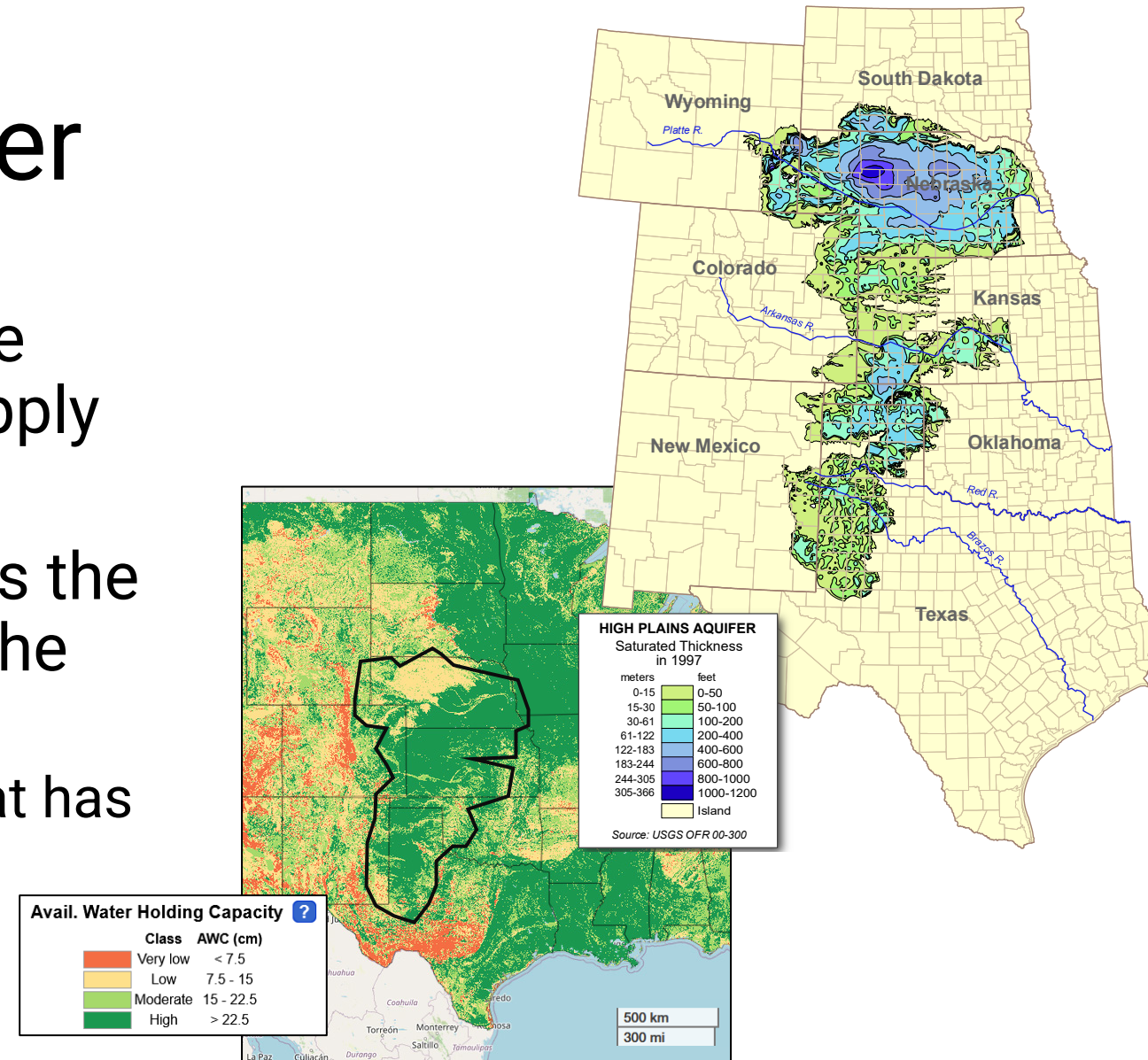


well-aggregated soils allow water to infiltrate, reducing the risk of flooding, and increasing the return of water to the aquifer.



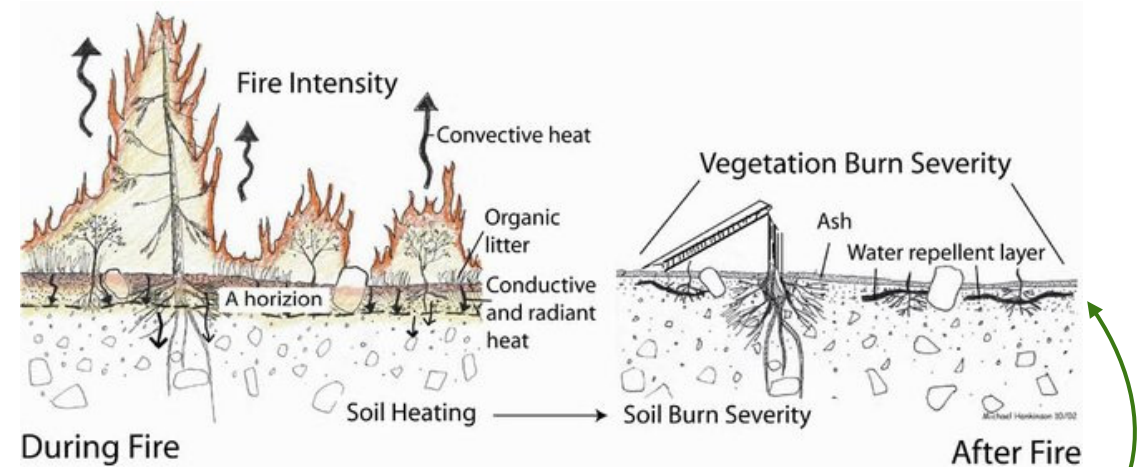
Shaping Groundwater

- Groundwater makes up a large amount of our fresh water supply (~49.3%)
- The largest Aquifer in the US is the **Ogallala Aquifer** (also called the High Plains Aquifer)
 - The Ogallala aligns with soil that has high **water holding capacity**



Flood Protection

- When soil is healthy, it can protect areas from flooding
 - This protection can be impacted by events like wildfires
- Soil scientists often map soil burn severity after wildfires
 - Early warning for erosion and flooding risk
 - Indicates potential for recovery



This **water-repellent layer** is dangerous since it keeps soil from absorbing water which can **increase flooding risk**

Soil is also more likely to **erode** after a fire which can be especially hazardous in the event of flooding

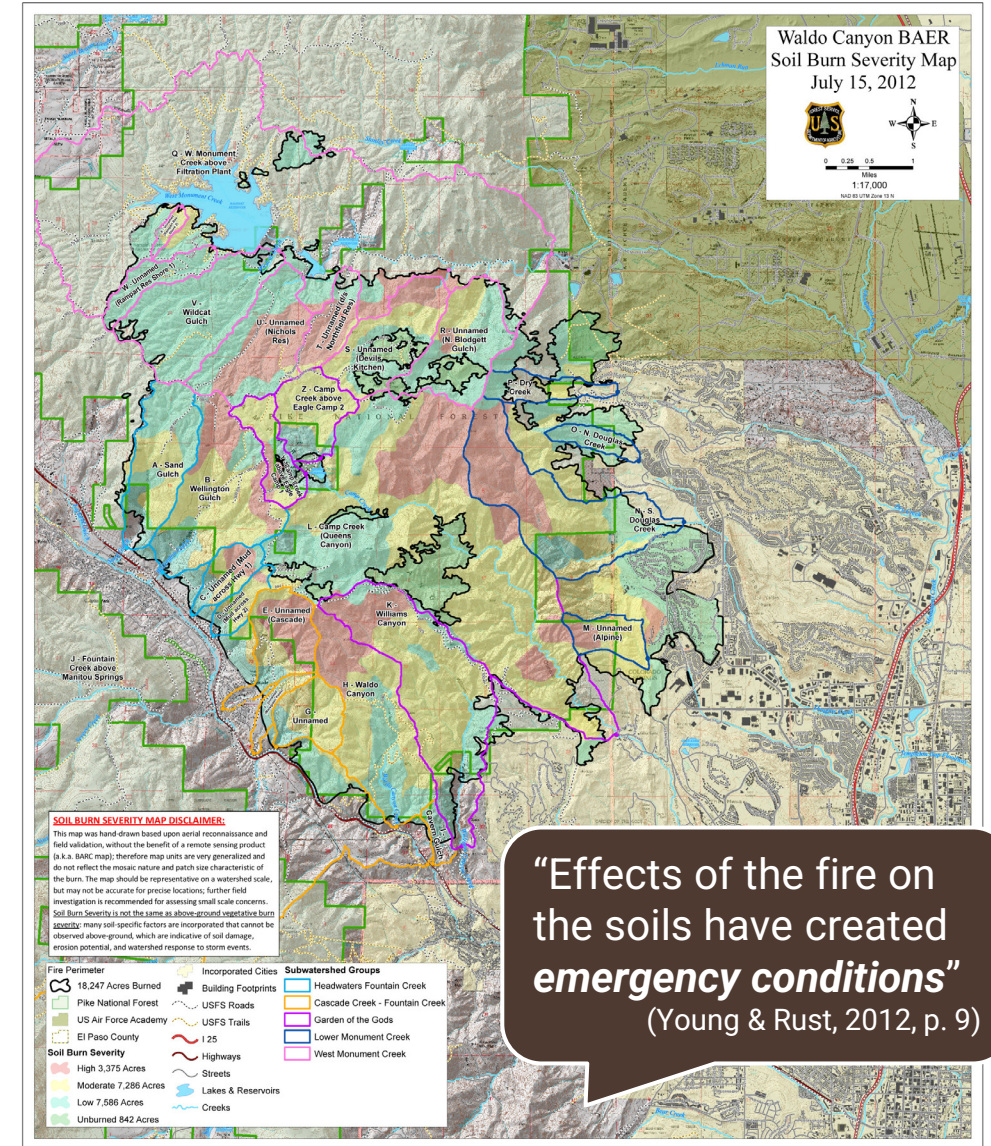


Waldo Canyon Fire (2012)

Burned June -
July 2012

- 18,247 acres
- Hundreds of homes lost

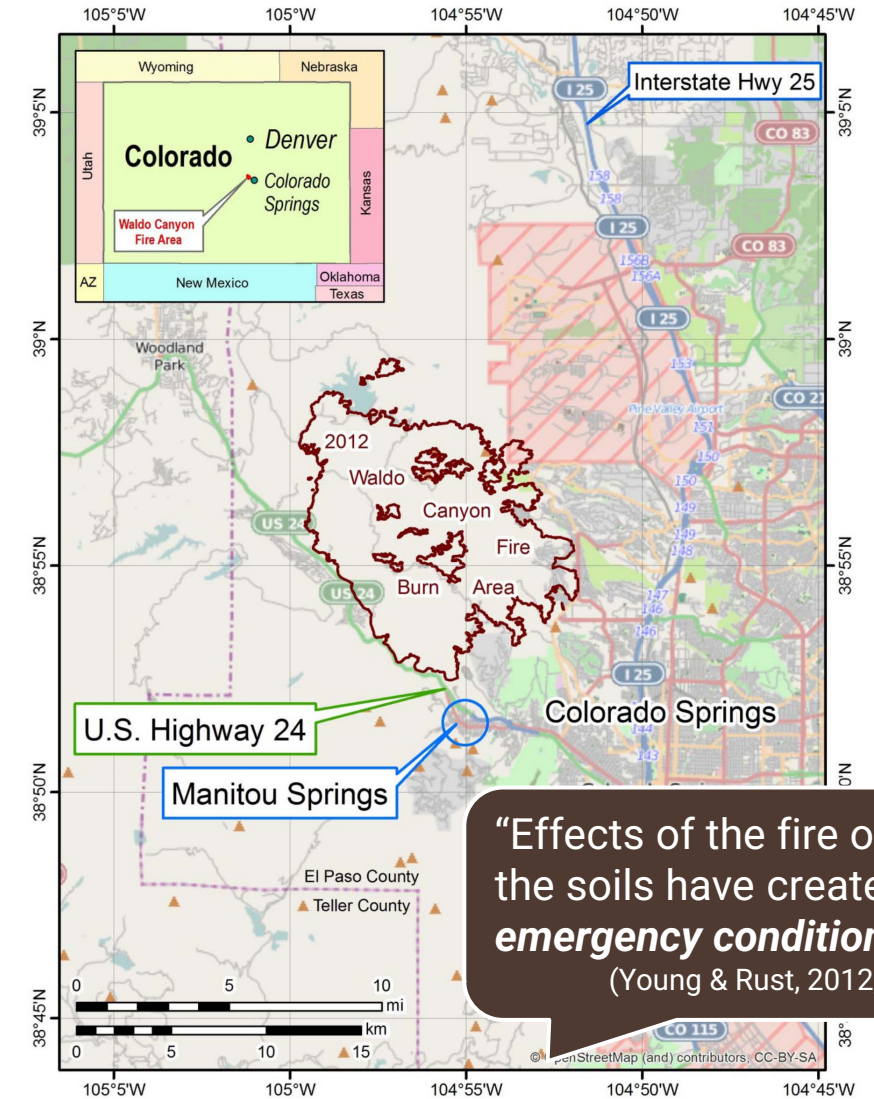
Most destructive
fire in Colorado
until the Black
Forest Fire in
2013



Manitou Floods (2013)

Only a year later,
flash floods hit
the region

The burned soil
was not able to
mitigate water
flow and was
especially prone
to erosion

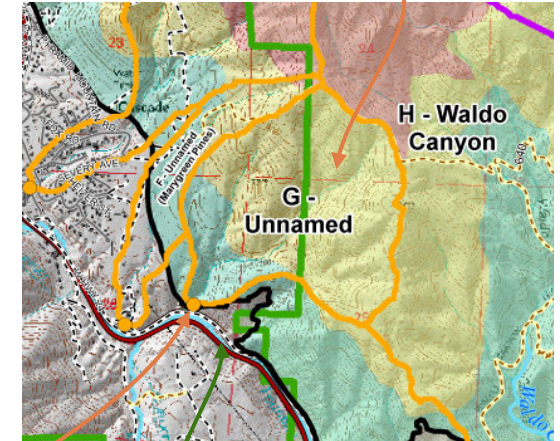


Manitou Floods (2013)

Unnamed Creek 1's drainage culvert to **Fountain Creek** was half-filled during the floods

- However, the sediment was free-draining **coarse sand**
 - This likely kept the water from overtopping US 24

Watershed of Unnamed Creek 1



Coarse Sand

Drainage Culvert from Unnamed Creek 1



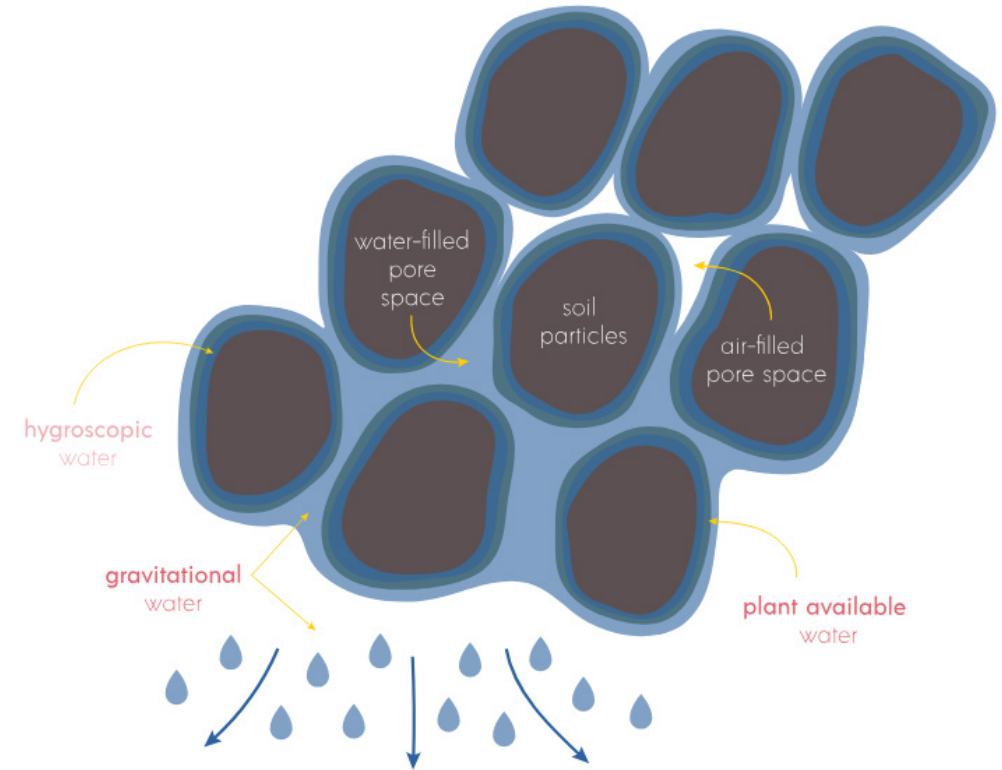
Fountain Creek



Soil Moisture

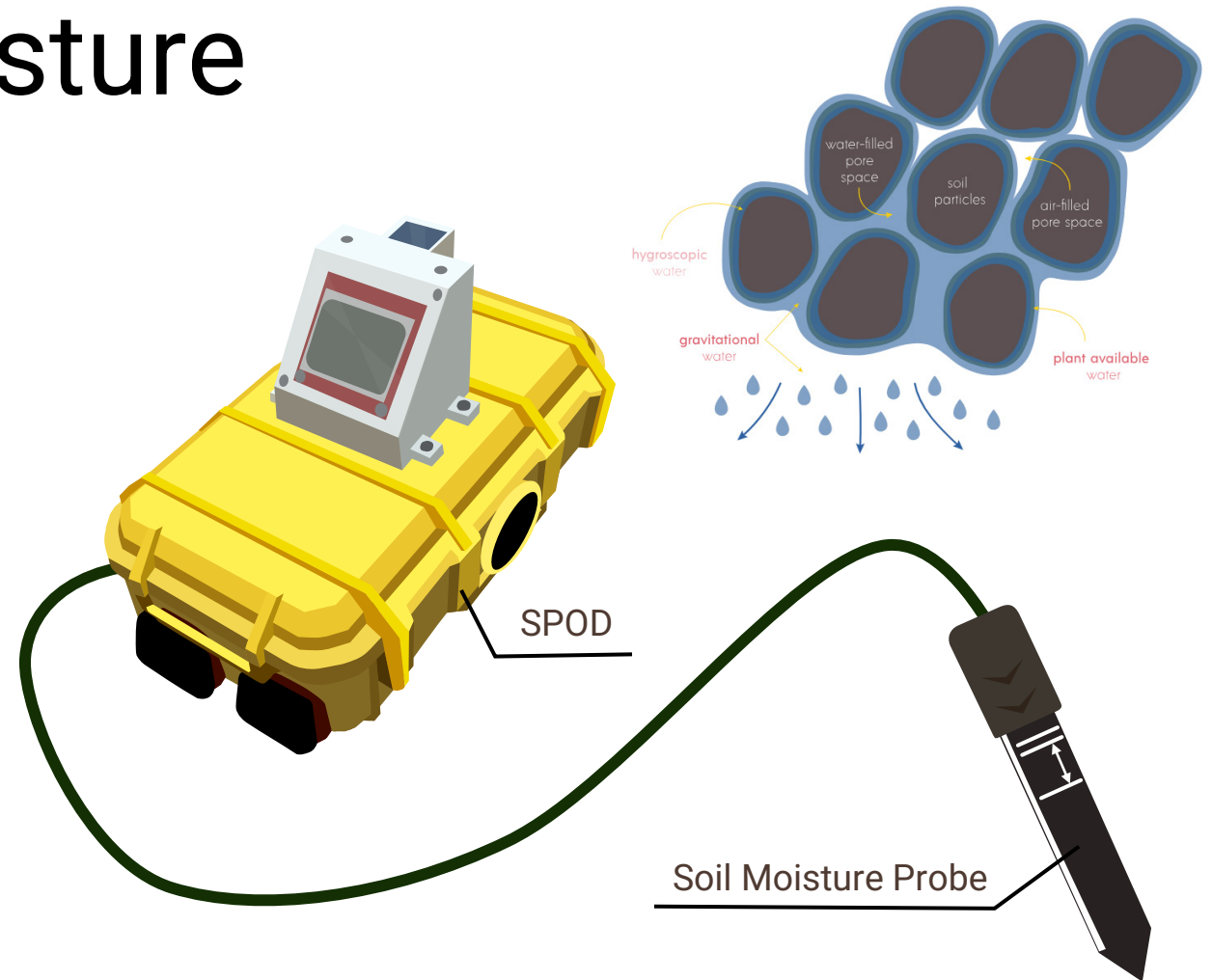
Soil Moisture tells us how much water is in the soil

- In the soil's pore space
- And in organic matter
 - Soil organic matter can hold 20x its weight in water!



Measuring Soil Moisture

- Let's return to the same samples with different questions:
 - Which sample currently has more water?
 - Which sample do you think could hold the most water?



How do we group, classify, and taxonomize soil?

What are characteristics of this soil?

What capacity does this soil have for this ecosystem?

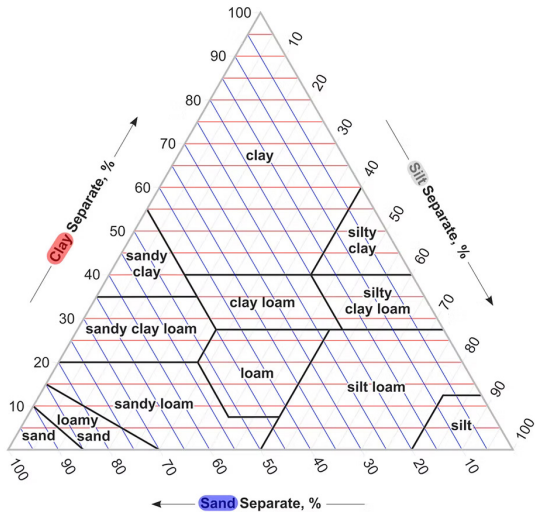
Soil Capability

Properties that relate to the soil's composition; these often go unchanged

Inherent Soil Qualities



Formation Factors

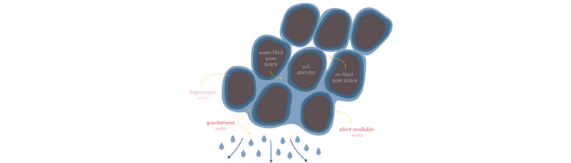


Soil Texture

Soil Health

Properties that change with use, management, and the environment

Dynamic Soil Qualities

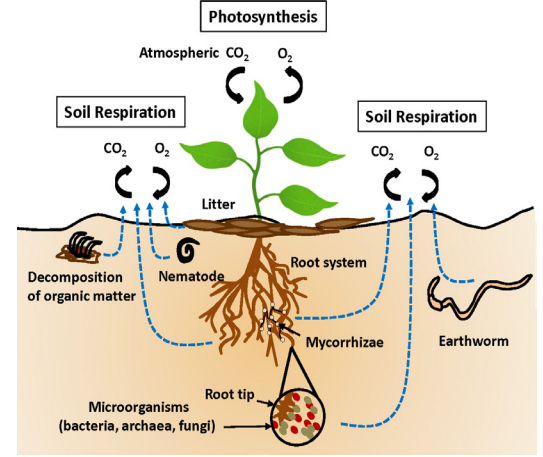


Soil Moisture

Is the soil able to function at its "full capacity"?

Can the soil function as a living ecosystem?

Can the soil sustain animals, plants, and humans?



Soil Respiration



If you want to learn more, we recommend starting with these excellent sources!

*Chiartas, J., Ward, A., Zahangir, K., Santih, S., Massenzio, E., Lew, S., Gomes, A., & Le, B. (2015). *Soil life*. UC Davis. Soil Life. <https://www.soillife.org/>*

Fortuna, A.-M. (2012). The Soil Biota. *Nature Education Knowledge*, 3(10), 1.
<https://www.nature.com/scitable/knowledge/library/the-soil-biota-84078125/>

Moebius-Clune, B. N. (2016). *Comprehensive assessment of soil health: The Cornell framework manual* (Third edition). Cornell University.
<https://soilhealth.cals.cornell.edu/manual/>

Needelman, B. (2013). What Are Soils? *Nature Education Knowledge*, 4(3), 2.
<https://www.nature.com/scitable/knowledge/library/what-are-soils-67647639/>

Walkinshaw, M., O'Green, A. T., & Beaudette, D. E. (2025). *Soil Properties* [Map]. California Soil Resources Lab. <https://casoilresource.lawr.ucdavis.edu/soil-properties/>

